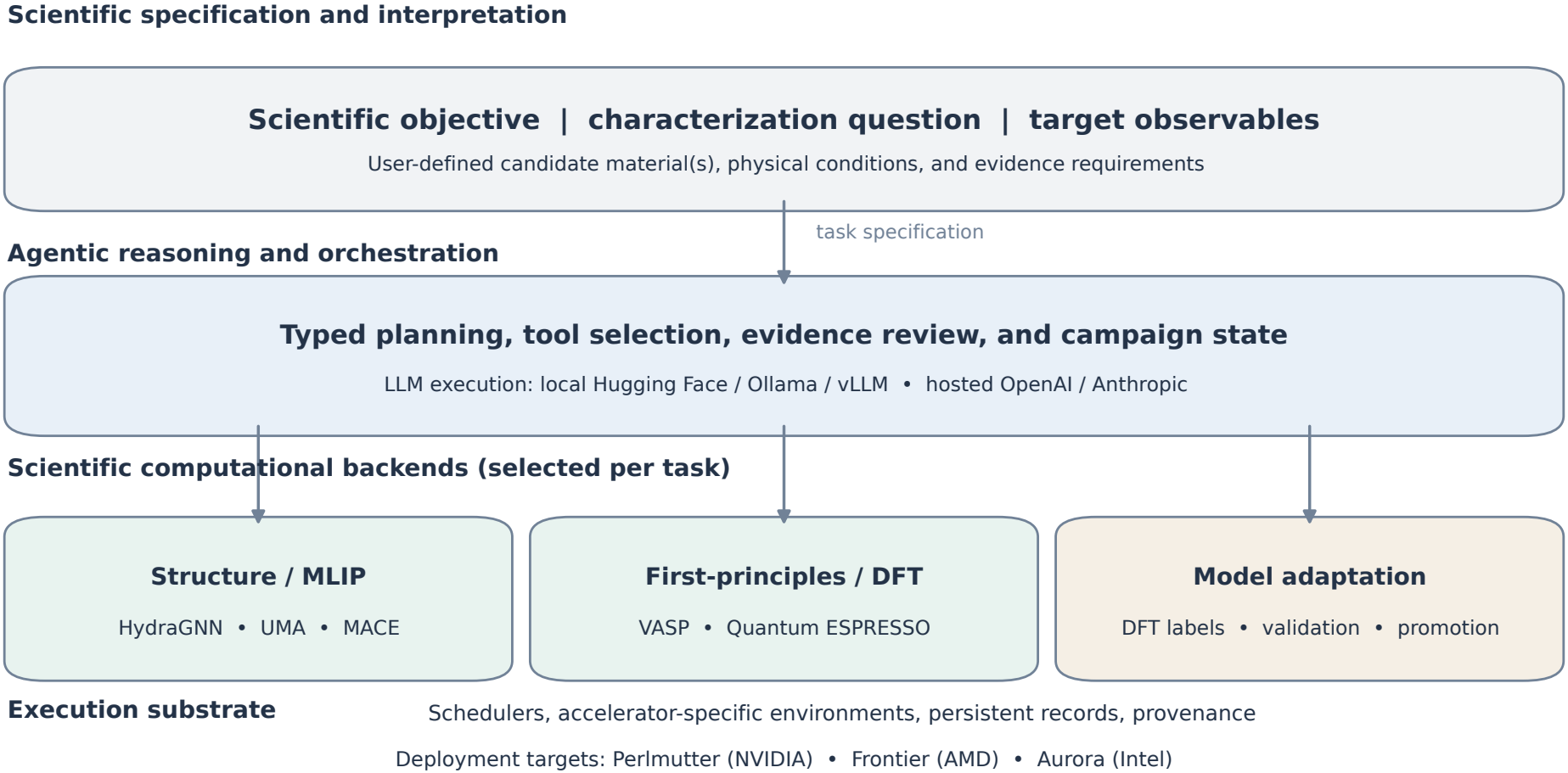
Scientific specification and interpretation
Scientific objective | characterization question | target observables
User-defined candidate material(s), physical conditions, and evidence requirements
task specification
Agentic reasoning and orchestration
Typed planning, tool selection, evidence review, and campaign state
LLM execution: local Hugging Face / Ollama / vLLM • hosted OpenAI / Anthropic
Scientific computational backends (selected per task)
Structure / MLIP
HydraGNN • UMA • MACE
First-principles / DFT
VASP • Quantum ESPRESSO
Model adaptation
DFT labels • validation • promotion
Execution substrate
Schedulers, accelerator-specific environments, persistent records, provenance
Deployment targets: Perlmutter (NVIDIA) • Frontier (AMD) • Aurora (Intel)